

Cover Letter

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Editor-in-Chief

Sensors

Dear Editor,

I am pleased to submit our manuscript entitled “**Efficient 1D Deep Learning Model on ECG for Multi-Class Atrial Arrhythmia and Flutter Subtype Classification**” for consideration as a **research article** in *Sensors*.

This study addresses the critical challenge of classifying atrial arrhythmias particularly the nuanced subtypes of atrial flutter (AFL) distinguished by specific atrioventricular (AV) conduction blocks—using single-lead ECG data. Differentiating such subtypes is essential for timely and accurate medical decisions, yet traditional approaches are time-consuming and require expert interpretation.

We contribute two major innovations:

1. A novel annotation method for ECG datasets that labels AFL subtypes with AV blocks from 1:1 to 4:1, enhancing clinical relevance.
2. A lightweight yet powerful deep learning model that utilizes depthwise separable convolutions, squeeze-and-excitation blocks, residual connections, and bidirectional gated recurrent units to classify arrhythmias directly from lead II ECG signals.

The proposed model demonstrates exceptional performance, achieving 98.23% accuracy, 98.25% precision, 98.29% recall, 98.27% F1 score, and 99.70% specificity. Its design is optimized for integration with wearable ECG devices, providing a viable path toward real-time monitoring and intelligent healthcare systems.

We believe this work is of strong interest to the readers of *Sensors* “MDPI” as it merges cutting-edge deep learning techniques with practical biomedical applications closely aligning with the journal's focus on innovative signal and image processing approaches for real-world impact.

- We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal.
- All authors have approved the manuscript and agree with its submission to *Sensors*.

Thank you for considering our manuscript for publication. We look forward to your feedback.

Sincerely,

Nadjem Eddine Menaceur

On behalf of all co-authors